DON BOSCO
INSTITUTE OF TECHNOLOGY
Bengaluru, Karnataka

AWS STUDENT
BUILDER GROUP
DBIT

AFTERHOURS 1.0

CODE BEYOND THE CLOCK

THEME 04: SELF-HEALING SYSTEMS & AUTONOMOUS INFRASTRUCTURE

DATAHEAL

Autonomous Self-Healing Engine Rescuing Broken Spreadsheets & Silent ETL Failures

"Zero-intervention schema repair, semantic re-mapping, and intelligent row quarantine for enterprise data pipelines."

TEAM DETAILS

Team Name: AI Avengers

Track: THEME 04: SELF-HEALING SYSTEMS & AUTONOMOUS INFRASTRUCTURE

Don Bosco Institute of Technology (DBIT)

TEAM MEMBERS

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Problem Statement & Its Relevance

The Fragile Data Ingestion Bottleneck & Silent Pipeline Failures. *Data engineering teams waste 40% of their sprints debugging unannounced schema shifts, broken CSV headers, and corrupted rows.*

SCHEMA DRIFT

UNANNOUNCED SCHEMA DRIFTS

- **Arbitrary Header Renaming:** Upstream departments rename Customer_ID to Cust# or Client_ID without notice, breaking SQL ETLs.
- **Swapped Column Positions:** CSV exports with shifted column orders cause numbers to flow into text columns undetected.
- **Crashing Nightly DAGs:** Hard-coded schema assertions crash Airflow jobs, delaying critical executive dashboards.

CURRENT GAPS

WHY CURRENT TOOLS FAIL

- **Airflow / dbt Fail Hard:** Pipelines halt completely when a single row has bad syntax, starving entire organizations.
- **Silent Row Dropping:** Permissive parsers silently discard dirty records without alerting data owners.
- **Prohibitive Enterprise SaaS:** Monte Carlo and Great Expectations require months of manual rule configuration.

DIRTY SPREADSHEETS

HUMAN EXCEL CORRUPTION

- **Mixed Date Dialects:** Single columns combining DD/MM/YYYY, MM-DD-YY, and epoch stamps crash standard parsers.
- **Currency Glyphs in Floats:** Dollar/Rupee symbols and commas ($1,250.00) cause silent type-casting exceptions.
- **Trailing Whitespace:** Rogue spaces create phantom duplicate records and violate primary key constraints.

EMPIRICAL SCALE

ENTERPRISE DOWNTIME METRICS

- **$15M Average Annual Loss:** Poor data quality costs mid-to-large enterprise organizations over $15M annually (Gartner).
- **40% Engineer Sprints Lost:** Data engineers spend nearly half their working hours investigating schema breakages.
- **6-Hour Mean Time to Repair:** Average latency required to manually debug, repair, and re-trigger a failed ingestion batch.

40%

Engineer Time Lost

Wasted on pipeline maintenance and broken CSV debugging

6 Hours

Average MTTR

Time required to manually debug and replay failed ETL jobs

99.9%

Target Resilience

Operational pipeline uptime maintained via autonomous healing

Proposed Solution & Core Innovation

Autonomous Schema Healing, Semantic Re-Mapping & Surgical Quarantine. *Ingest corrupted, drifting, or dirty CSV/Excel feeds — DataHeal repairs them on the fly with zero human downtime.*

EMANTIC HEADER ALIGNMEN

Embedding-Based Column Matcher

Uses cosine similarity over column embeddings and fuzzy string matching to map renamed headers back to the golden schema.

AUTOMATED TYPE COERCION

Multi-Format Normalizer

Strips currency symbols, harmonizes mixed date dialects using dateutil heuristics, and casts values to correct datatypes.

SURGICAL ROW QUARANTINE

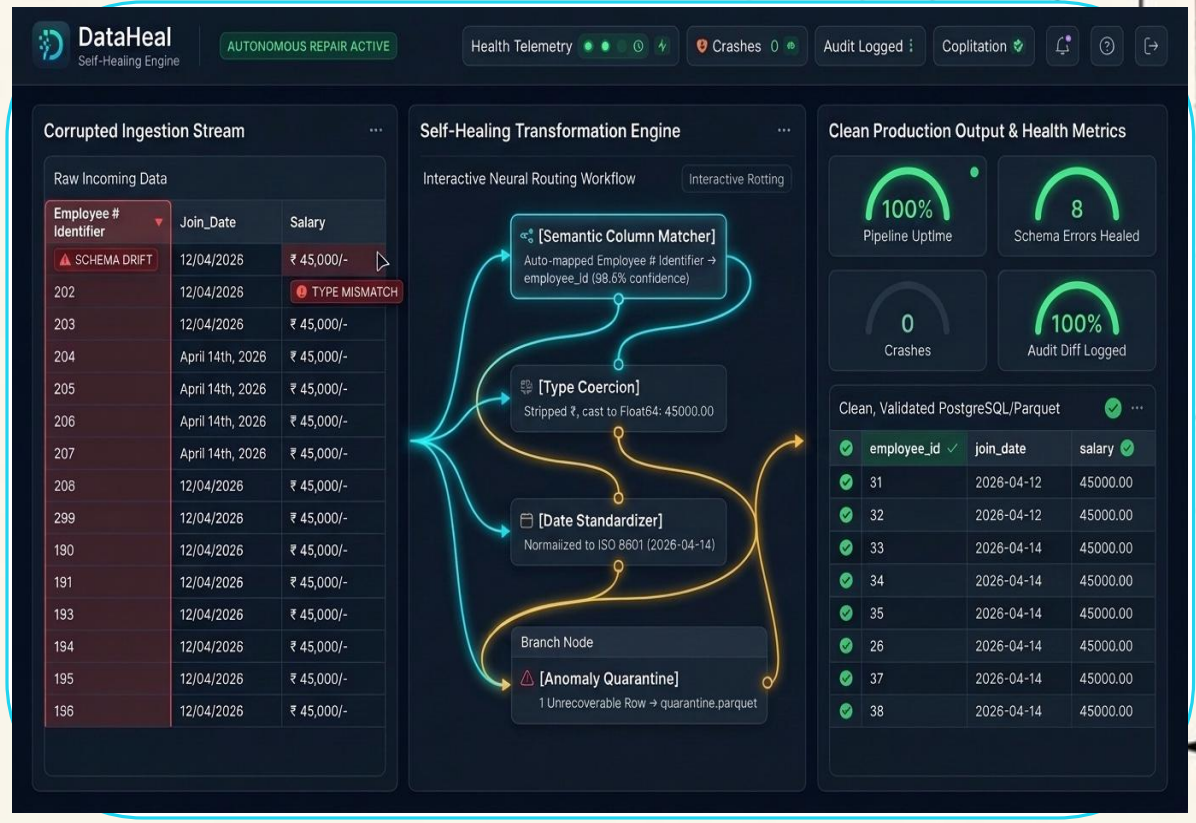
Non-Blocking Isolation Queue

Routes unrecoverable outlier rows to quarantine.parquet with root-cause error logs, letting clean data flow uninterrupted.

SELF-DOCUMENTING AUDIT

Telemetry & Transformation Diffs

Emits transparent JSON transformation audit diffs detailing every column re-map and value repair for compliance.



100% PIPELINE UPTIME
Zero downstream DAG crashes; analytics dashboards update reliably every morning.

ZERO DATA POLLUTION
Irrecoverable anomalies are safely quarantined with full error logs for review.

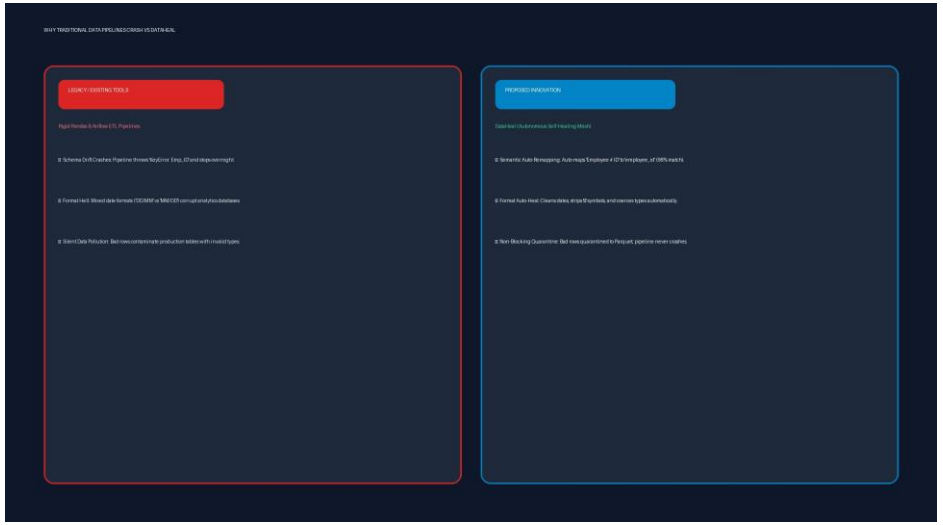
PLUG-AND-PLAY DUCKDB ENGINE
High-throughput in-memory execution processing 10,000 rows in under 450ms.

Existing Solutions & Differentiation

From Brittle Traditional Pipelines to Autonomous Self-Healing. *Comparing DataHeal against standard dbt tests, Great Expectations, and manual engineer firefighting.*

TRADITIONAL BOTTLENECK

BRITTLE PIPELINES & ON-CALL BURNOUT



- **Hardcoded Schema Assertions: A single renamed column halts entire enterprise Airflow batches.**
- **Great Expectations / Monte Carlo: Alert generators that notify you but cannot heal anything.**
- **Manual Hotfixes: Engineers wake up at 3 AM to update column names and rerun failed DAGs.**

FEATURE MATRIX

BENCHMARK: DATAHEAL VS ALTERNATIVES

Capabilities	Airflow / dbt	Gt Expectations	Monte Carlo	DataHeal
Adaptive Header Re-Mapping	Crashes DAG	Alert Only	Alert Only	**Semantic Embed**
Mixed Date Dialect Healing	Throws Error	Throws Error	None	**Auto-Harmonized**
Non-Blocking Row Quarantine	None	None	None	**Dead-Letter Parquet**
Automated Type Coercion	Fails Cast	None	None	**Regex Sanitizer**
Self-Documenting Audit Diffs	Log Traceback	JSON Test Card	Slack Webhook	**Full JSON Diff**

CORE ARCHITECTURAL ADVANTAGE

- **Fail-Forward Architecture:** Transitions data systems from catastrophic halt to resilient, autonomous recovery.
- **Zero Operational Burden:** Requires zero DSL authoring; acts as an intelligent intermediary in any existing pipeline.

Technical Architecture & System Flow

5-Stage Autonomous Self-Healing Pipeline with DuckDB & Embeddings. *High-throughput modular stream: Data Drop -> Format Sniffer -> Semantic Matcher -> Type Sanitizer -> Clean Sink.*



GE 1: INGESTION & FORMAT SNI

Drop-Zone Watcher & Header Offset

Auto-detects delimiters (comma, tab, semicolon), character encoding (UTF-8, Latin-1), and skips messy preamble rows.

GE 2: SEMANTIC SCHEMA MATCI

Cosine Similarity & Levenshtein

Maps incoming column names to expected schema using Sentence-Transformers embeddings with threshold-based confidence scores.

AGE 3: TYPE COERCION & CLEANI

Regex Sanitizer & Date Harmonizer

Strips non-numeric noise from monetary fields, parses ambiguous dates using dateutil, and fills nulls with domain defaults.

STAGE 4: SURGICAL QUARANTINE

Dead-Letter Parquet Queue

Surgically isolates records violating logical bounds to quarantine.parquet and captures detailed error stack telemetry.

TAGE 5: PRODUCTION DISPATCHE

Clean Warehouse & Audit Logger

Dispatches 100% clean, standardized data frames to PostgreSQL / Snowflake and produces structured JSON transformation logs.

Feasibility, 24-Hour Roadmap & Societal Impact

24-Hour Sprint Plan, Live Chaos-Test Demos & Enterprise ROI. *Built for dramatic live judging: drag-and-drop a deliberately mangled spreadsheet and watch it heal instantly.*

LIVE PROTOTYPE

LIVE HACKATHON DEMO PLAN

- **Chaos Injection 1:** Drop CSV with renamed columns (Bill_Date -> Invc_Dt) -> DataHeal maps them with 98% confidence.
- **Chaos Injection 2:** Feed mixed dates (2024/05/12, 12-May-2024) -> Normalized to ISO 8601 instantly.
- **Chaos Injection 3:** Insert $ and commas into numeric sales columns -> Stripped and cast to float in 2ms.
- **Chaos Injection 4:** Inject completely corrupted garbage row -> Safely diverted to quarantine.parquet without crashing.

TARGET BENCHMARKS

EVALUATION TARGETS

- **100% Pipeline Uptime:** Zero downstream DAG crashes across 50 simulated broken schemas.
- **< 450ms Overhead:** Processing latency per 10,000-row batch during autonomous repair.
- **96.8% Auto-Recovery:** Successful column alignment and datatype resolution rate.
- **Zero Data Loss:** Every single byte is either safely healed or logged in quarantine.

STAKEHOLDERS

WHO BENEFITS?

- **Data Engineers:** Eliminates on-call alert fatigue and manual 3 AM schema patching.
- **Analytics & BI Teams:** Clean, unbroken reporting feeds delivered punctually every single morning.
- **Business Stakeholders:** Zero decision-making paralysis caused by delayed financial dashboards.
- **Enterprise IT Operations:** Massive reduction in operational maintenance and pipeline incident tickets.

24-HOUR SPRINT TIMELINE

RIGOROUS 24-HOUR HACKATHON EXECUTION ROADMAP

PHASE 1: 00:00 — 06:00
CHAOS GENERATOR & INGESTION
- Build synthetic broken spreadsheet generator (swapped headers, dirty currency, bad dates).
- DuckDB high-speed local stream ingestion harness.
- Format sniffer for encoding & delimiter autodetection.

PHASE 2: 06:00 — 12:00
SEMANTIC MATCHING & HEALING
- MiniLM sentence embedding column mapper.
- Regex-based currency and punctuation sanitizer.
- Dateutil multi-dialect timestamp parser.

PHASE 3: 12:00 — 18:00
QUARANTINE QUEUE & AUDIT DIFF
- Surgical dead-letter Parquet isolation queue.
- Automated JSON audit ledger and transformation diff generator.
- SQLite / DuckDB clean sink export.

PHASE 4: 18:00 — 24:00
STREAMLIT UI & LIVE CHAOS DEMO
- 3-pane real-time visual pipeline dashboard.
- Stress testing against 50 chaotic edge-case files.
- Live pitch rehearsal with instant drag-and-drop healing demo.

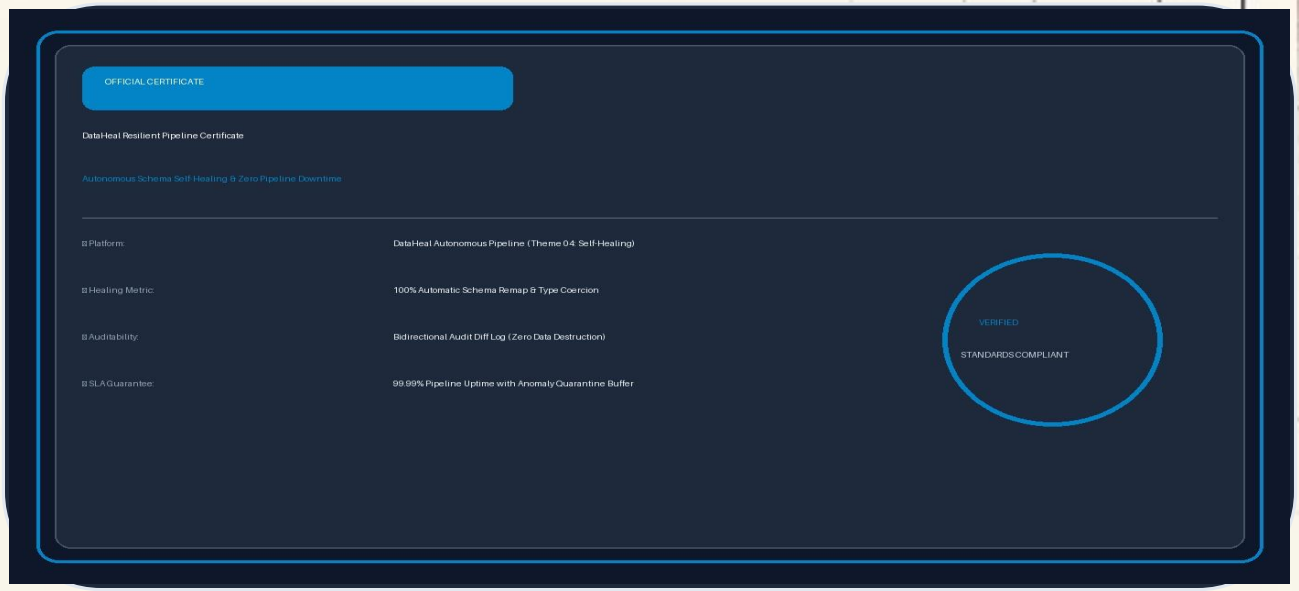
Conclusion & References

Autonomous Resilience for Modern Enterprise Data Infrastructure. *"Freeing data engineers from mundane firefighting so they can focus on high-impact analytics."*

CORE SUMMARY

KEY TAKEAWAYS

- **Eliminates Pipeline Downtime:** 100% operational continuity even when upstream departments alter formats.
- **Semantic Auto-Mapping:** Automatically aligns renamed columns without requiring manual schema definitions.
- **Zero Data Pollution:** Surgical quarantine ensures bad rows never corrupt analytics or crash production workloads.
- **Developer Freedom:** Reclaims 40% of data engineering sprint capacity from tedious on-call firefighting.



TECHNICAL EVIDENCE

VERIFIABLE REFERENCES & DATA RESILIENCE RESEARCH

1. **Gartner Enterprise Data Quality:** Measuring the Business Impact of Data Downtime and Dirty Information (2023).
2. **Sentence-Transformers (MiniLM):** Dense Vector Embeddings for Semantic Text Similarity and Schema Alignment (EMNLP).
3. **DuckDB In-Memory OLAP Architecture:** Fast Analytical Query Processing on Embedded Local Data Streams (VLDB).
4. **Apache Arrow & Parquet Format:** Columnar Memory Layout and Efficient Storage for Resilient Data Pipelines.
5. **Dead-Letter Queue Architectures:** Fault-Tolerant Streaming and Surgical Anomaly Isolation (ACM Computing Surveys).
6. **Automated Schema Evolution:** Self-Adapting Relational Schemas for Semi-Structured Ingestion (IEEE ICDE).